

Big Data Infrastructure, Weather Prediction Using Weather APIs

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ABSTRACT

This project will implement a real-time weather prediction system streaming data from APIs such as Weatherstack using Apache Kafka to realize on-the-spot forecasting and warning of significant events with regard to weather impacts, such as storms and visibility issues. Furthermore, Prometheus is applied to ensure the stability of the real-time data pipeline through metric collections. The system follows real-time monitoring and batch processing with the help of Apache Hadoop or Spark for volume data in historical weather records. This batch and dual-stream approach enables short-term prediction and long-term trend analysis. Predictive models will be developed using Apache Spark MLlib, which would permit precise forecasting from the historical data. It stores the results of the processed batch in HDFS, scaling access to insights that power data-driven decisions related to weather planning and response.

Keywords: Weather Prediction System Historical Data Analysis, Spark MLlib, Distributed File Storage (HDFS), Grafana, Prometheus.

1 INTRODUCTION

Weather forecasting has become very important in various sectors for better management and the safety of citizens, especially in cities like Cape Town, which experience high climatic variability. This

system of weather prediction focuses on real-time updates with long-term forecasting to cater to various sections such as agriculture, tourism, and disaster management. It will utilize real-time weather data from streams such as the Weatherstack API and make it available through Apache Kafka so that instantaneous and timely forecasts can be done for storms, extreme temperatures, or visibility. In addition, historical trend analysis of the weather through batch processing will be able to predict much more accurate future forecasts. The Prometheus was also used to gather and store system performance and weather metrics as time-series data that could be graphically represented in Grafana for real-time dashboards and alerts.

There are two major focal areas of this project: one is the delivery of the current weather in real time, and the other is using past weather to carry out predictive modeling. In this respect, real-time data will enable the system to track current conditions and send warnings for urgent weather events, whereas batch processing through Apache Hadoop and Apache Spark will provide the processing of long-term historical weather trends. With this fusion, therefore, the system will be able to predict short-term weather patterns but also show recurring patterns that hint at broader climate behavior in Cape Town. This approach ensures preparedness now and the accuracy of the forecast in the longer term. Furthermore, Prometheus enables real-time monitoring whereby it collects weather data along with system metrics that could also be analyzed

for immediate decision-making.

Some of the key integrated technologies include the Weatherstack API for live weather data, Apache Kafka for real-time streaming, Apache Spark for processing large volumes of data, and MLlib for predictive analytics. Hadoop can provide a distributed file system to store huge amounts of historical weather data, whereas Spark's MLlib is used to build machine learning models predictive of weather patterns [1]. For the same forecasting and performance of the system, Grafana offers an easy-to-use dashboard for the rendering of the same weather data and operational metrics. Prometheus and Grafana together provide the essential monitoring and visualization layer to track the health of the system for easy analysis of weather patterns that affect agriculture, tourism, and disaster management sectors in Cape Town



IN PICTURES | Strong winds tear through Cape Town

Figure 2: Enter Caption

2 SYSTEM ARCHITECTURE



Figure 1: figure 1: Showing Poor visibility in Cape Town road N2.

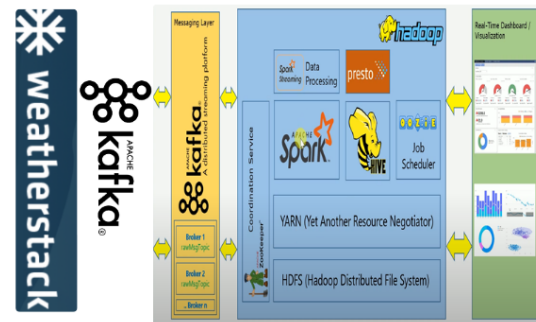


Figure 1: System Architecture Design for Weather Prediction

The architecture is robust; the system processes historical and real-time weather data to predict hazardous weather conditions in Cape Town while ensuring that data flow from acquisition to visualization.

Data Acquisition and Streaming Real-time weather data of wind speed, visibility, and temperature, among other weather-related metrics, are gathered

from Weatherstack API. This information is fetched through a Kafka Producer, which acts as the data source sending the weather information into a Kafka Topic. Then, the Kafka Consumer gets this real-time data from the topic and lets the continuous stream of weather information. This real-time data stream is treated with Apache Spark, doing transformations, computations, and aggregations required. The processed data is then stored in HDFS for long-term storage and further analysis.

Historical Data and Predictive Modeling Along with the real-time data, historic weather data is also collected from sources like Kaggle. This data, several years of weather record keeping, is also persisted in HDFS. The system trains a Linear Regression Model on this historic dataset using MLlib of Apache Spar (see [1] and [2]). It is now supposed to predict, based on major variables such as wind speed and visibility, hazardous weather conditions in Cape Town. These are, in fact, vital conditions towards the identification of extreme weather, which has considerable impacts on public safety, agriculture, and tourism. Consequently, the model will make a forecast of the future weather conditions after training, while the forecasted data derived thereof is preprocessed and stored for visualization.

Data Visualization and Monitoring The result of the Linear Regression model for predicted weather data and the real-time data coming from Kafka will be visualized to provide easy access and further analysis. The system will utilize Prometheus for fetching time-series data, both in real-time and predicted weather metrics and in system performance metrics. Prometheus continuously monitors such metrics; it stores them in a time-series database for efficient retrieval [3]. The data is then relayed to Grafana, which displays the current weather data from Kafka and the forecasted weather conditions based on the regression model output. Grafana allows intuitive graphical representation of real-time weather in Cape Town, thus the response to emerging weather threats within agriculture, tourism, and disaster management will be facilitated by ease [4].

3 REAL-TIME WEATHER DATA STREAM

The system retrieves real-time weather data from the Weatherstack API, which provides up-to-date information about temperature, humidity, wind speed, visibility, and storm alerts for Cape Town. This data is essential for providing accurate and timely weather forecasts. The system queries the Weatherstack API at regular intervals, ensuring that the weather data is constantly refreshed. Key parameters such as temperature, wind speed, visibility, and humidity are critical for predicting immediate weather events like storms, heavy winds, and reduced visibility, which can significantly impact daily life and operations in Cape Town.

For this flood of real-time data, the system uses Apache Kafka, a distributed event streaming platform. Kafka can efficiently transport the weather data coming from Weatherstack API to the consuming services of the system. This flow of data takes place in a pre-structured Kafka topic, where it gets published by a Kafka Producer and gets consumed by one or more Kafka Consumers. For example, topics can be created for holding general weather data, storm warnings, or visibility problems in each case. It is guaranteed by Kafka that it will handle weather data in a fault-tolerant and real-time fashion so that the system could instantly process it and react on the changing conditions.

Kafka integrated into the system makes it scalable and fault-tolerant to work with big streams of weather data. The Kafka Producer collects data about the weather from Weatherstack and sends it into an appropriate Kafka topic. While the Kafka Consumer subscribes to the topics, processing them in real time. Further, this information will be passed on to a processing system like Apache Spark for deep analysis or to trigger alerts based on the predetermined conditions of this data. Kafka provides those infrastructural provisions necessary to handle the continual streaming of data, which keeps it scalable as the amount of weather data grows over time while being collected and processed.

4 Batch Data Processing

For the completed weather prediction system, historical weather data plays a key role in long-term trend analysis and predictive model development. This data is stored and managed efficiently using the Hadoop Distributed File System (HDFS), which allows the system to handle large datasets generated over extended periods. Apache Hadoop and Apache Spark are used to process the data, utilizing MapReduce or Spark's distributed computing power to analyze and extract valuable insights from historical weather patterns. By processing this data, the system can uncover important trends like seasonal shifts, anomalies, and extreme weather events, providing a deeper understanding of Cape Town's weather.

Using Apache Spark MLlib, the system trains predictive models to forecast weather conditions such as wind speed and visibility, two critical factors for safety. The historical data undergoes preprocessing steps like feature extraction, normalization, and transformation to ensure it is suitable for model training. Once the model is trained, it is evaluated for performance and accuracy, ensuring it can reliably predict dangerous weather conditions. These models, based on long-term trend analysis, allow the system to issue alerts and provide real-time forecasts, helping stakeholders in sectors like agriculture, tourism, and disaster management to make informed decisions and take timely action based on predictive weather data.

5 Predictive Weather Model, Visibility and Windspeed

In this weather prediction project, two Linear Regression models were built to predict visibility and wind speed based on weather-related features such as temperature, humidity, and pressure. The models were trained using data stored in a CSV file within HDFS (Hadoop Distributed File System). The goal was to forecast visibility in kilometers and wind speed in kilometers per hour. Data preprocessing steps like feature scaling were performed using VectorAssembler and StandardScaler to ensure the models performed optimally.

The first model, the Visibility Model, predicted visibility based on temperature, humidity, and pressure. The coefficients of the model revealed how each feature affected visibility. Temperature had a positive effect on visibility, while humidity negatively impacted it. Pressure had a smaller positive influence. The model's intercept of 10.34 represented the baseline visibility when all features were zero, which is used as a reference for prediction.

The second model, the Wind Speed Model, predicted wind speed based on the same weather features. In this case, all features had a negative coefficient, meaning higher temperature, humidity, and pressure were associated with lower wind speeds. The intercept for this model was 10.82, again representing the baseline wind speed. The prediction results for both models were evaluated using Root Mean Squared Error (RMSE) to determine their accuracy.



Figure 1: Visualisation of regression prediction performance

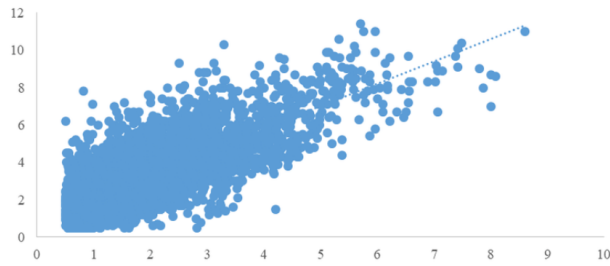


Figure 1: WindSpeed of regression prediction performance

The performance of the models was measured using RMSE, which quantifies the average deviation between predicted and actual values. The Visibility Model had an RMSE of 3.79, suggesting it was fairly accurate in predicting visibility. On the other hand, the Wind Speed Model had a slightly higher RMSE of 6.57, indicating that it was less precise in forecasting wind speed compared to the visibility model.

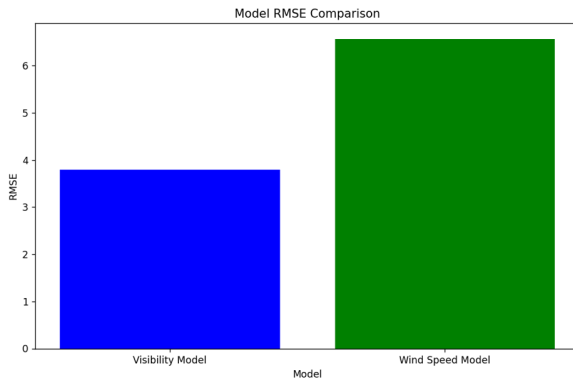


Figure 3: Visualisation of the Model Performance of Visibility and WindSpeed

Both models demonstrated reasonable performance, with the Visibility Model outperforming the Wind Speed Model in terms of accuracy. These models can be further refined by incorporating additional weather features or using more sophisticated algorithms to improve their predictive capabilities. By storing the models in HDFS, they are available for future weather predictions, providing valuable insights for

forecasting visibility and wind speed in real-time weather applications.

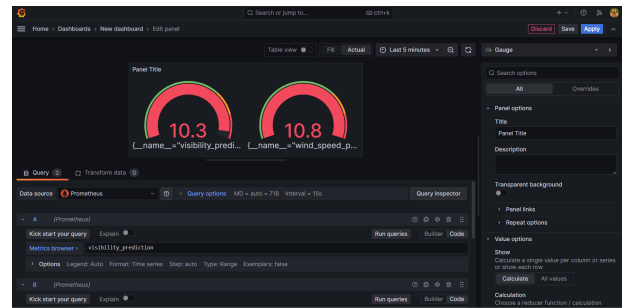


Figure 3: Figure: Visualisations of Visibility and WindSpeed on grafana

In the project, the predictive model is specifically designed to generate real-time alerts for changes in visibility and wind speed. By analyzing incoming weather data, the model predicts significant shifts in visibility or wind speed that may indicate hazardous conditions. When the model detects a threshold that could potentially lead to unsafe weather conditions, such as a sudden reduction in visibility or an increase in wind speed, it triggers an alert. These alerts are based on the model's predictions and aim to provide timely warnings to ensure safety.

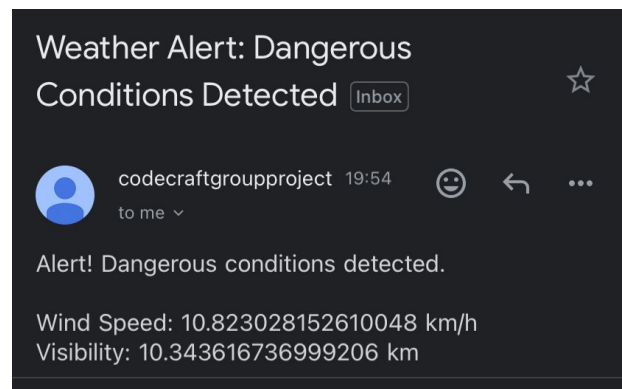


Figure 4: Figure : Showing Alerts Sent on Email for dangerous Weather Conditions

Once the model generates an alert, it is distributed through multiple channels to ensure maximum reach.

Alerts regarding visibility and wind speed changes are communicated via email, SMS, or mobile app notifications, allowing relevant stakeholders, such as emergency responders, travelers, or local authorities, to take the necessary precautions. This system ensures that users are promptly informed of adverse weather conditions, improving response times and helping to prevent potential accidents or disruptions caused by sudden changes in visibility or wind speed.

6 CONCLUSION

The project has successfully designed a weather forecasting and alerting system for real-time prediction and long-term analysis, using machine learning models to predict critical parameters like visibility and wind speed. Thus, it ensures timely warnings against hazardous conditions by leveraging the historical weather data stored in HDFS and applying predictive modeling. In the future, this may include the integration of more feeds of weather data, optimization of predictive models to achieve higher accuracy, and the addition of more personalized features in the alert system. This system thus has immense potential for application in agriculture, tourism, and Public safety within Cape Town, enabling informed decisions, enhancing safety, and generally improving the levels of preparedness in response to any changes in weather conditions.

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